
Question 1.1. Let's run a hypothesis test using confidence intervals to see if there is a linear relationship between egg weight and bird weight. Define the null and alternative hypotheses that will allow you to conduct this test. **(8 points)**

Note: Please write your answer **in the cell below** in the following format: - **Null Hypothesis:** - **Alternative Hypothesis:**

Hint: Reminder your null hypothesis should be about the relationship in the population.

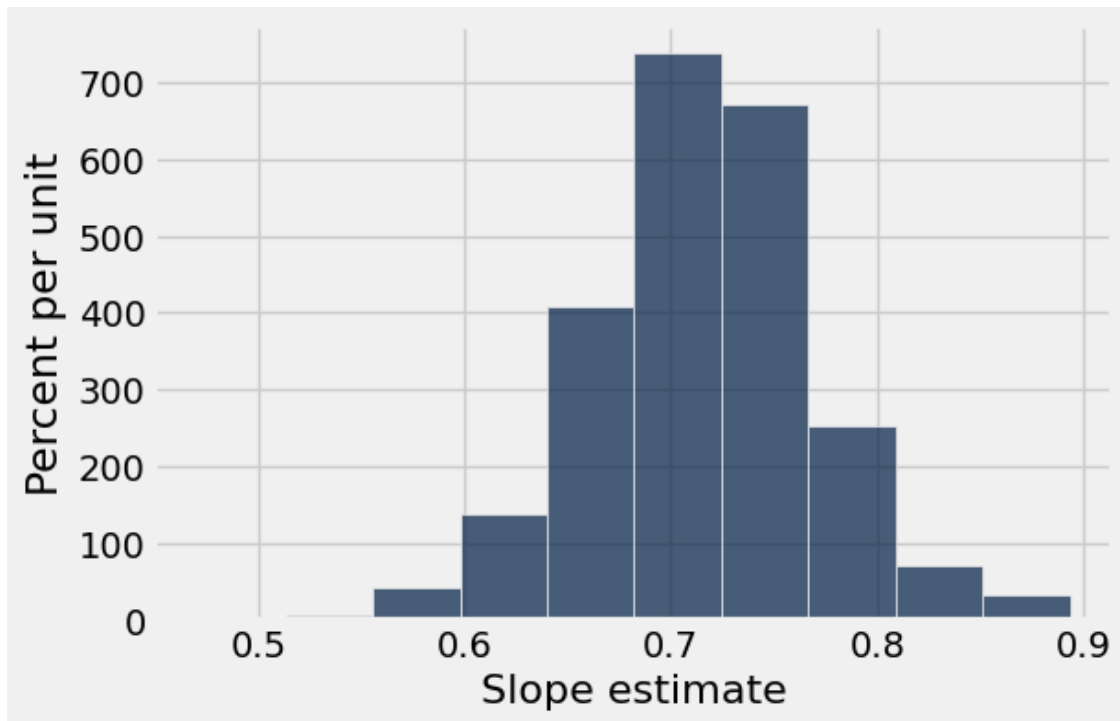
Null Hypo: There is no relationship between egg weight and bird weight for Snowy Plovers in the population
Alternative Hypo: There is linear relationship between egg weight and bird weight for Snowy Plovers in the population

Question 1.4. Create an array called `resampled_slopes` that contains the slope of the best fit line for 1000 bootstrap resamples of `birds`. Plot the distribution of these slopes. (8 points)

```
In [14]: resampled_slopes = make_array()

for i in np.arange(1000):
    birds_bootstrap = birds.sample(with_replacement = True)
    bootstrap_line = fit_line(birds_bootstrap, "Egg Weight", "Bird Weight")
    bootstrap_slope = bootstrap_line.item(0)
    resampled_slopes = np.append(resampled_slopes, bootstrap_slope)

# DO NOT CHANGE THIS LINE
Table().with_column("Slope estimate", resampled_slopes).hist()
```



```
In [15]: grader.check("q1_4")
```

```
Out[15]: q1_4 results: All test cases passed!
```

Question 1.6. Based on your confidence interval, would you reject or fail to reject the null hypothesis that the true slope is 0? Why? What p-value cutoff are you using? **(8 points)**

Hint: Read the introduction of this homework!

reject null hypothesis because I assume that p-value is 0.05 $\Rightarrow$ then I need to find 95% confidence interval of the slope

$\Rightarrow$ from the result above: 95% confidence interval for slope: [0.60261, 0.823017]

but 0 is not in that interval $\Rightarrow$ reject null hypothesis

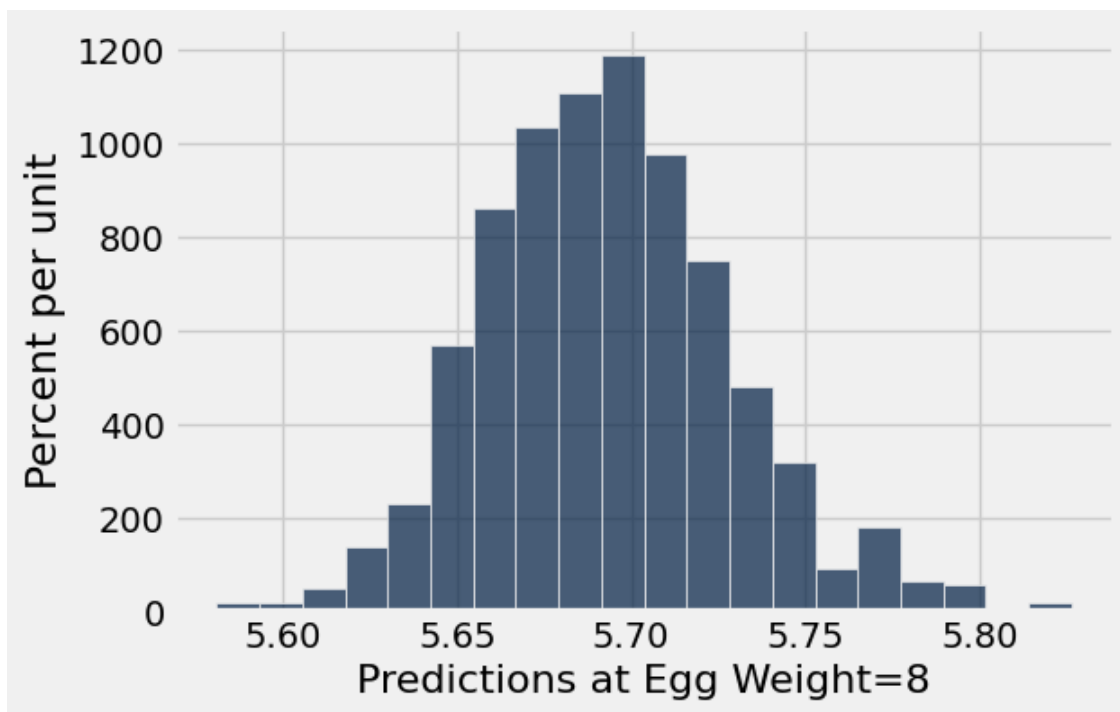
Question 1.7. Using your results from previous questions, **provide an estimated range for the true slope**. How confident are you that the true slope falls within this range? **(8 points)**

Estimated range for the true slope: $[0.60261, 0.823017]$

Confident that the true slope is in the range: 95%

Question 2.4. Create an array called `predictions_for_eight` that contains the predicted bird weights based on an egg of weight 8 grams for each regression line in `regression_lines`. (6 points)

```
In [34]: predictions_for_eight = np.array([8 * slope + intercept for slope, intercept in zip(regression_lines,
# This will make a histogram of your predictions:
table_of_predictions = Table().with_column('Predictions at Egg Weight=8', predictions_for_eight)
table_of_predictions.hist('Predictions at Egg Weight=8', bins=20)
```



```
In [ ]: grader.check("q2_4")
```

